

Complete Backend Development Topics (2025 In-Demand Roadmap)

● Level 1 – Foundations (Beginner Must-Haves)

Goal: Build strong backend basics and APIs.

1. How the Web Works

- Client–Server Model
- DNS, IP, Ports, HTTP/HTTPS
- Request–Response lifecycle
- REST vs GraphQL vs gRPC vs WebSockets

2. JavaScript for Backend

- Async Programming (Callbacks, Promises, Async/Await, Event Loop)
- Modules (CommonJS, ES6 imports/exports)
- Error Handling (try/catch)
- OOP & Prototypes in JS

3. Node.js Fundamentals

- What is Node.js (V8, single-threaded, event-driven)
- Core Modules (fs, path, http, crypto)
- npm & package.json
- EventEmitter
- Streams & Buffers

4. Express.js (Core Web Framework)

- Creating APIs
- Routing (GET, POST, PUT, DELETE)
- Middleware (built-in, custom, third-party)
- Error handling middleware
- Static file serving & file uploads (Multer)

5. APIs & HTTP

- RESTful API design principles (CRUD, Pagination, Sorting, Filtering)
- Status Codes (200, 201, 400, 401, 403, 404, 500)
- Postman / Thunder Client API testing

6. Mini Project 1 → Task Manager API (Express, in-memory storage, CRUD)

● Level 2 – Databases, Auth & Real-World APIs

Goal: Connect backend to databases, add auth, and make production-ready APIs.

7. Databases

- SQL → PostgreSQL/MySQL
 - Joins, Indexing, Transactions, ACID properties
- NoSQL → MongoDB (Mongoose)
 - Schemas, Models, Aggregation, Indexing, Sharding
- Redis (Caching, Sessions, Pub/Sub)
- CAP Theorem

8. Authentication & Authorization

- Sessions & Cookies
- JWT (JSON Web Tokens)
- OAuth 2.0 & OpenID Connect (Google/GitHub login)
- RBAC (Role-Based Access Control)
- MFA (Multi-Factor Authentication)

9. GraphQL & gRPC

- GraphQL Queries, Mutations, Subscriptions (Apollo Server)
- gRPC basics (Protocol Buffers, high-performance RPCs)

10. WebSockets (Real-Time Apps)

- Socket.IO with Node.js (chat, notifications, live tracking)

11. Security Basics

- SQL/NoSQL Injection
- XSS, CSRF attacks & protection
- HTTPS & SSL/TLS
- Helmet.js, Rate limiting

12. Mini Project 2 → User Auth System

- Register/Login with JWT + MongoDB
- Role-based access (Admin vs User)

● Level 3 – Advanced Backend (Industry Demand)

Goal: Become production-ready with advanced scaling, security, DevOps.

13. Advanced Node.js & Performance

- Clustering & Worker Threads
- Load Balancing (Nginx, HAProxy)
- Caching strategies (Redis, CDN)
- Logging (Winston, Morgan)

14. Microservices Architecture

- Monolith vs Microservices
- API Gateway (Express Gateway, Kong, Nginx)
- Service Discovery
- Inter-service communication (REST, gRPC, Message Queues)

15. Event-Driven Systems

- Pub/Sub pattern
- Message Brokers → Kafka, RabbitMQ, SQS
- Event Sourcing & CQRS basics

16. Cloud & DevOps Basics

- Git & GitHub (Version Control)
- CI/CD pipelines (GitHub Actions, Jenkins, GitLab CI)
- Cloud deployment (AWS, GCP, Azure)
- Docker (Containerization)
- Kubernetes (Orchestration)
- Infrastructure as Code (Terraform basics)

17. Security (Advanced)

- OAuth 2.0 deep dive
- Secure password storage (bcrypt, Argon2)
- API security best practices
- Secrets management (dotenv, Vault)

18. System Design & Scalability

- High-Level Design: load balancers, CDNs, caching, sharding
- Low-Level Design: designing scalable APIs
- Consistency Models (Strong, Eventual)
- CAP Theorem in practice

19. Monitoring & Observability

- Logging (ELK stack: Elasticsearch, Logstash, Kibana)
- Metrics (Prometheus + Grafana)
- Error tracking (Sentry)

20. Mini Project 3 → E-Commerce Backend

- Products, Orders, Payments (Stripe API)
 - Authentication + Authorization
 - Caching with Redis
 - Deployment with Docker/K8s
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Bonus (Future-Proof Skills)

- **Serverless** (AWS Lambda, Vercel Functions, Netlify)
- **Domain-Driven Design (DDD)**
- **Clean Architecture & SOLID principles**
- **Polyglot exposure** → Learn basics of Go/Java/Python backend to increase versatility
- **AI-Integrated Backends** → Serve ML models with Node.js APIs